Umair Yousaf

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Education

University of Colorado Boulder, MS in Computer Science - CGPA 4.0 Aug 2025 – May 2027

- **Courses:** Distributed Systems, Intro. to Robotics, Transformers for Robotics, Deep Reinforcement Learning, Neuro-Symbolic NLP, Linear Programming

Lahore University of Management Sciences (LUMS), BS in Computer Science - CGPA 3.6 Aug 2019 – May 2023

Technologies

Languages: Python, C++, JavaScript, SQL
Robotics: ROS2, NVBlox, Open3D, PCL, RViz2, RTDE, UR5e, Franka Pandas, Clearpath Ridgeback, Intel RealSense, RGB-D Perception
ML / AI: PyTorch, TensorFlow, HuggingFace, SAM 3, LangChain, LangGraph, YOLO, Vision Transformers, RAG
Cloud & Tools: Azure (ML, Container Instances, DevOps), Docker, gRPC, Flask, Git

Experience

University of Colorado Boulder, HIRO Lab, Graduate Researcher - Boulder, CO May 2026 – Present

- Built a real-time perception pipeline on **Intel RealSense** RGB-D input, combining **SAM 3** instance segmentation with **NVBlox** volumetric 3D reconstruction for object detection; derived per-object surface normals from the reconstruction to compute perpendicular approach poses for contact-rich interaction.
- Deployed the **Compliant Explicit Reference Governor (CERG)** on a dual-arm platform (**2x UR5e** arms on a **Clearpath Ridgeback**) hosted at **PickNik Robotics**, enforcing bounded interaction forces during robot-environment contact; tuned PD gains and validated dual-arm joint-goal commanding. NASA-funded collaborative robotics, targeting crew-safe robotic assistance on lunar space stations.
- Enabled joint-level **torque/effort control** on the UR5e arms by upgrading `ur_robot_driver`, modifying **RTDE** recipe files, and updating xacro hardware interfaces; maintained platform uptime through PolyScope upgrades and a full arm replacement following an encoder failure.

NSF Institute for Student-AI Teaming, Graduate Researcher - Boulder, CO Feb 2026 – Present

- Developing an AI-assisted classroom discourse monitoring system for student collaborative learning, applying **neuro-symbolic NLP** techniques with **LLaMA 3.3 70B** for real-time utterance classification and dialogue management.
- Built a similarity-based dialogue router using **cosine embeddings** to classify and route student utterances, reducing LLM misclassification; migrated vector store from ChromaDB to **Qdrant** for improved retrieval performance.
- Integrated LLMs with real-time speech detection using **Whisper** and **SileroVAD** for noisy classrooms; system deployed and evaluated across live classroom pilots in K–12 settings.

Reteta, Junior Consultant (AI Team) - Cranbury, NJ · Remote Nov 2024 – Jul 2025

- Fine-tuned LLMs on patient-doctor interaction data to generate structured SOAP notes with **95% accuracy**, reducing physician documentation time by **75%**.
- Deployed and maintained a scalable ML backend on **Azure** using Azure ML and Azure Container Instances; built CI/CD pipelines with **Azure DevOps** for automated testing, deployment, and monitoring.
- Developed **NER**-based PII anonymization pipeline ensuring HIPAA compliance across all patient data workflows.

Visionet Systems, Junior Consultant (AI Team) - Lahore, Pakistan Jul 2023 – Nov 2024

- Developed a PDF parsing pipeline integrated with **Agentic AI and RAG** architectures, improving document retrieval efficiency by **20%**; pipeline extracted text, tables, images, captions, headers, and footers across large insurance corpora.
- Built end-to-end Generative AI solutions for insurance underwriting and claim management workflows using **LangChain** and **LangGraph**, streamlining client operations.

Aletheia-AI (now BrickAndMortar.AI), ML & SWE Intern - Lahore, Pakistan Jun 2022 – Aug 2022

- Deployed real-time detection systems using **YOLOv7** for vehicle, ANPR, and person detection, achieving **>90% detection accuracy** in production.

Projects

LEGAL-UQA: Low-Resource Urdu-English Legal QA Dataset | **NLP/ML** | **arXiv preprint**

- Created the first Urdu-English legal QA dataset (619 Q&A pairs) from Pakistan's constitution via OCR extraction and **GPT-4**-assisted translation; evaluated generalist LLMs and embedding models, reaching **99.19% human-evaluated accuracy** with Claude-3.5-Sonnet.